Ginzburg and then Doroshkevich, Novikov, and Zel'dovich discover the first evidence that a black hole has no "hair." [Ch. 7]

Colgate, May, and White in the U.S., and Podurets, Imshennik, and Nadezhin in the U.S.S.R., adapt bomb design computer codes to simulate realistic implosions of stellar cores; they confirm Zwicky's 1934 speculation that implosions with low mass will form a neutron star and trigger a supernova, and confirm the 1939 Oppenheimer–Snyder conclusion that implosions with larger mass will create a black hole. [Ch. 6]

Zel'dovich, Guseinov, and Salpeter make the first proposals for how to search for black holes in the real Universe. [Ch. 8]

Salpeter and Zel'dovich speculate (correctly) that supermassive black holes power quasars and radio galaxies. [Ch. 9]

Herbert Friedman and his team discover Cygnus X-1, using a Geiger counter flown on a rocket. [Ch. 8]

**1965** Boyer and Lindquist, Carter, and Penrose discover that Kerr's solution of Einstein's field equation describes a spinning black hole. [Ch. 7]

**1966** Zel'dovich and Novikov propose searching for black holes in binaries where one object emits X-rays and the other light; this method will succeed in the 1970s (probably). [Ch. 8]

Geroch shows that the topology of space can change (for example, a worm-hole can form) non–quantum mechanically only if a time machine is created in the process, at least momentarily. [Ch. 14]

**1967** Wheeler coins the name *black hole.* [Ch. 7]

Israel proves rigorously the first piece of the black-hole, no-hair conjecture: A nonspinning black hole must be precisely spherical. [Ch. 7]